

E04 CSTR Energy Balance Example

The elementary liquid phase reaction $A \rightarrow B$ is carried out in a non-isothermal CSTR. Species A and an inert I are fed to the reactor in equimolar amounts. The molar feed rate of A is 80 mol/min.

The heat capacity of the inert is 30 cal/mol/°C. The heat capacities of A and B are 20 cal/mol/°C.

$$\tau = 100 \text{ min}$$

$$\Delta H^{rxn} = -7500 \text{ cal/mol}$$

$$UA = 8000 \text{ cal/min/}^\circ\text{C}.$$

$$T_a = 300 \text{ K}$$

$$k = 0.0066 \text{ min}^{-1} \text{ at } 350 \text{ K}$$

$$E = 40 \text{ kcal/mol}.$$

Determine:

- The reactor temperature for a feed temperature of 450 K.
- The ignition temperature.
- The extinction temperature.

Decision Guide: Non-Isothermal CSTR Energy Balance Example

Goal

Determine:

1. The reactor temperature when the feed temperature is 450 K.
2. The ignition temperature.
3. The extinction temperature.

Step 1: Identify the Type of Problem

Ask yourself:

What reactor?

→ CSTR

Is temperature specified?

→ No

Is heat transfer present?

→ Yes (UA and T_a are provided)

This is a: Non-Isothermal CSTR with a heat exchanger.

The solution will require:

- A CSTR mole balance
- A CSTR energy balance
- Temperature-dependent kinetics

Step 2: Build the Energy Balance First

The problem provides:

- UA
- T_a

- Feed temperature
- Heat capacities
- Heat of reaction

These are clues that the energy balance will be central to the solution.

For a CSTR with heat transfer, use:

$$X_{EB} = \frac{\frac{UA}{F_{A0}}(T - T_a) + \sum_i \Theta_i C_{Pi}(T - T_0)}{-\Delta H_{rxn}^{\circ}}$$

CBE 3610 Reactor Design Handbook Equation (28)

Step 3: Calculate the Heat-Capacity Term

Determine:

$$C_{P0} = \sum_i \Theta_i C_{Pi}$$

using:

- A and I are equimolar in the feed.
- C_{PA}
- C_{PI}

This quantity is needed in the energy balance.

Step 4: Create the Energy-Balance Relationship

Use Equation (28) or the simplified CSTR form:

$$X_{EB} = \frac{C_{P0}(1 + \kappa)(T - T_c)}{-\Delta H_{rxn}^{\circ}}$$

where

$$\kappa = \frac{UA}{F_{A0} C_{P0}}$$

Equation (29)

and

$$T_c = \frac{\kappa T_a + T_0}{1 + \kappa}$$

Equation (30)

giving

$$X_{EB}(T)$$

Equation (31)

Step 5: Write the Reactor Kinetics

The problem states: Elementary reaction: $A \rightarrow B$

Therefore:

$$-r_A = kC_A$$

using the general rate-law framework.

CBE 3610 Reactor Design Handbook Equations (44)–(45)

Step 6: Account for Temperature Dependence

The problem provides:

- Activation energy
- Reference rate constant
- Reference temperature

Use:

$$k(T) = k(T_{ref}) \left[\frac{E}{R} \left(\frac{1}{T_{ref}} - \frac{1}{T} \right) \right]$$

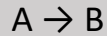
Equation (49)

This gives:

$$k = f(T)$$

Step 7: Write Concentration in Terms of Conversion

For:



under liquid-phase conditions:

$$C_A = C_{A0}(1 - X)$$

Use this to express:

$$-r_A = f(X, T)$$

Step 8: Write the CSTR Mole Balance

Use:

$$V = \frac{F_{A0}X}{(-r_A)}$$

CBE 3610 Reactor Design Handbook Equation (21)

Substitute:

- $k(T)$
- $C_A(X)$

to obtain:

$$X_{MB}(T)$$

Step 9: Create the Mole-Balance Curve

You now have:

Energy Balance:
 $XEB(T)$

Mole Balance:
 $XMB(T)$

Both equations are functions of temperature.

Step 10: Find the Reactor Temperature

Plot:

$XEB(T)$

and

$XMB(T)$

on the same graph.

The intersection points represent valid steady-state solutions.

For:

$T_0 = 450 \text{ K}$

identify the operating steady state.

Step 11: Determine Ignition and Extinction Temperatures

Use the same graphical framework.

Recall:

Ignition Temperature = feed temperature at which the lower steady state disappears.

Extinction Temperature = feed temperature at which the upper steady state disappears.

See the definitions in the Handbook. To determine these:

1. Vary feed temperature.
2. Recalculate the energy-balance curve.
3. Identify the temperatures where the curves become tangent.

These correspond to ignition and extinction conditions.

Final Answer Checklist

Before submitting your solution, verify that you have:

- ✓ Calculated C_{P0}
- ✓ Written the non-isothermal CSTR energy balance using Eq. (28) or Eq. (31)
- ✓ Calculated $k(T)$ using Eq. (49)
- ✓ Written the mole balance using Eq. (21)
- ✓ Developed $X_{EB}(T)$
- ✓ Developed $X_{Mb}(T)$
- ✓ Identified the operating temperature for $T_0 = 450$ K
- ✓ Determined the ignition temperature
- ✓ Determined the extinction temperature

CBE 3610 Reactor Design Handbook Equations Needed

Eq. (21) CSTR Mole Balance

Eq. (28) CSTR Energy Balance

Eq. (29) κ Definition

Eq. (30) T_c Definition

Eq. (31) Simplified CSTR Energy Balance

Eq. (44) General Rate Law

Eq. (45) Power-Law Kinetics

Eq. (49) Arrhenius Equation

Arrhenius Equation



$k(T)$

Mole Balance



$XMB(T)$

Energy Balance



$XEB(T)$

Intersection



Steady-State Temperature



Ignition / Extinction Analysis

This problem is fundamentally a **steady-state multiplicity problem** in which the reactor temperature is determined by the balance between **heat generated by reaction** and **heat removed by cooling and sensible-heat effects**.